

FoRAGe: High-CTR Food Image Synthesis with Retrieval-Augmented Diffusion Model

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Abstract

High Click-Through Rate (CTR) imagery has proven commercial value for food delivery platforms, driving a need for strategies to generate visually compelling images. Our investigations reveal a positive correlation between appropriate food backgrounds and subsequent user engagement. Despite advancements in diffusion models, inpainting new backgrounds does not guarantee high CTR, and fine-tuning diffusion models for this purpose is prohibitively expensive for the fast-paced online food delivery advertising sector. Consequently, there is a lack of cost-effective, transferable generation frameworks tailored to high-CTR food images. In this paper, we propose **FoRAGe**, a novel high-CTR *Food image Retrieval-Augmented Generation* pipeline leveraging ControlNet based on Stable Diffusion. Specifically, we construct a comprehensive food image database encompassing a diverse range of background environments. During image generation, FoRAGe retrieves high-quality

background exemplars featuring analogous food subjects from the database and employs the retrieved backgrounds as conditions to guide image synthesis via the ControlNet model. Subsequently, a multimodal CTR prediction model is utilized to identify and select optimal images for deployment. Extensive online experiments demonstrate a significant increase in CTR for images generated by our proposed pipeline, and ablation studies further elucidate the impact of different strategies and configurations. Code is available at <https://github.com/jiaxu-feng/FoRAGe>.

CCS Concepts

• **Applied computing** → **Online shopping**; • **Information systems** → **Online advertising**; • **Computing methodologies** → **Computer vision**.

Keywords

Click-Through Rate Prediction; Retrieval-Augmented Generation; Diffusion Model; Food Delivery

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Figure 1: Examples of food items with varying backgrounds. The CTR is displayed below each image. Higher values highlighted in red indicate increased consumer engagement.

1 Introduction

Ele.me, the online food delivery and local services platform of the Alibaba Group, is one of the largest in China. It serves hundreds of millions of customers and sellers alike. For E-commerce platforms like Ele.me, display advertising constitutes a crucial revenue stream, with a primary goal being the maximization of the Click-Through Rate (CTR). Existing research [1, 7] underscores the influence of visual appearance on user engagement. Additionally, CTR prediction models leveraging advanced architectures for visual feature extraction have been effectively deployed on E-commerce platforms, yielding substantial improvements in CTR [25, 43]. Our findings also reveal the significant impact of image backgrounds on CTR within the food delivery domain. We specifically address food delivery advertising across multiple platforms, including the Ele.me App, as well as WeChat and Alipay mini-programs. These advertisements predominantly appear in prominent vertical poster formats, targeting users on both the homepage and various category pages. As illustrated in Figure 1, the backgrounds of food images are critical in drawing customer attention. However, many sellers, particularly small restaurants, often lack the resources to produce images with well-composed backgrounds, causing plain backgrounds to diminish the appeal of high-quality food items. Thus, our objective is to generate new backgrounds for food images while preserving the integrity of the food subject’s pixels.

In recent years, the field of AI-Generated Content (AIGC) has advanced rapidly, significantly transforming various domains, including image generation. A key task within this domain is text-to-image (T2I) generation, where diffusion-based methods [17] currently represent the state-of-the-art approach. Models such as Stable Diffusion have demonstrated exceptional capabilities in creating high-quality images based on human prompts. Nonetheless, several challenges persist in tailoring these methods for specific applications, such as food image generation. Firstly, while diffusion-based

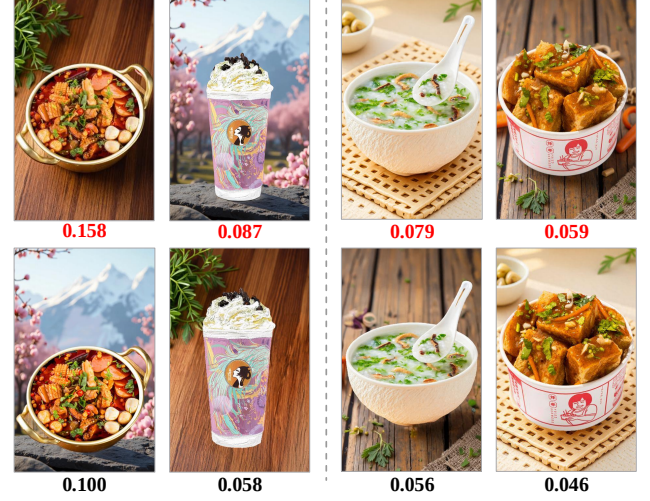


Figure 2: Examples of background swapping. The first row displays images with contextually harmonious generated backgrounds, while the second row shows the same items with swapped backgrounds. CTR scores are below the images.

methods are proficient in generating high-quality backgrounds, these backgrounds are not necessarily positively correlated with CTR [44]. Our online experiments, as depicted in Figure 2, underscore that merely employing high-quality backgrounds does not ensure an increase in CTR. Instead, the food content must be aptly matched with a background that is consistent in color, style, and thematic content to maximize user engagement and CTR. Secondly, fine-tuning diffusion models to optimize CTR might incur prohibitively expensive computational costs. Moreover, rapidly shifting consumer preferences demand a framework that supports fast iteration, balances efficiency and performance, and is suitable for industrial-scale deployment. Therefore, there is a noticeable absence of cost-effective transferrable generation frameworks specifically designed for food images, which is crucial for the fast-moving online food delivery advertising sector. Finally, despite extensive advances in image synthesis, no prior work has addressed the dual-fidelity challenge of generating new images in which (1) the primary subject remains unaltered to satisfy commercial-compliance requirements and (2) a high-quality reference background is maintained with minimal stylistic variation to preserve output quality. Therefore, the challenge extends beyond producing aesthetically compelling images; it also requires preserving high fidelity of both the food subject and the background, while ensuring their seamless integration.

To address the aforementioned challenges, in this paper, we introduce a novel pipeline, termed **FORAGE** (**F**ood image **R**etrieval-Augmented **G**eneration). Our methodology comprises a three-stage process. First, we perform a retrieval step, where high-quality images featuring food subjects semantically similar to the target image are retrieved from a meticulously curated database containing a diverse collection of food images with various backgrounds. The semantic similarity is quantified using a pre-trained feature extractor to encode both the target image and the database images into

a shared embedding space for image retrieval. Then, the retrieved images serve as references for a ControlNet-based generator to synthesize k candidate images, enabling background transplantation onto the target image. Lastly, to mitigate the issue of generating images with low CTR, we train a CTR prediction model, fine-tuned on food-specific datasets, to evaluate and select the most effective image from the generated candidates for deployment.

The contributions of this paper are summarized as follows:

- We propose a novel pipeline named **FoRAGE**, which integrates ControlNet-based RAG and a multimodal CTR prediction model for high-CTR food image synthesis.
- We are the first to deploy a CTR-optimized AIGC framework for customer-facing commercial scenarios on food delivery platforms, which is also adaptable to other domains.
- We introduce a new task termed Hi-Fi (**H**igh-**F**idelity) Background Replacement, defined as image-driven background replacement with stylistic consistency and little background variations, guided by a similar reference image.
- We have deployed the framework on the Ele.me food delivery platform. Results of online A/B tests and ablation studies demonstrate the pipeline’s superior performance in real-world scenarios.

2 Related Work

2.1 Image Synthesis

Image synthesis represents a pivotal task within the domain of computer vision, focusing on the creation of realistic and visually appealing images either from scratch or by modifying existing images. The evolution of frameworks such as Generative Adversarial Networks (GAN) [12], Variational Autoencoders (VAE) [19], and diffusion models [17] has significantly advanced the methodologies used in image synthesis. Currently, the Stable Diffusion family has emerged as one of the most prominent models for image generation. Within this field, food images have garnered considerable attention from researchers as a specialized domain, offering unique challenges and opportunities for exploration in image synthesis tasks. Early works predominantly employed GANs to generate food images [13, 31, 32, 47]. For example, CookGAN [13] uses an attention-based ingredients–image association model for synthesizing meal images from ingredients. More recently, diffusion models have emerged as a powerful alternative, such as Foodfusion [36] and ChefFusion [23]. Foodfusion employs a diffusion model augmented with a Fusion Module to integrate foreground and background information in food image composition. However, the paper does not address CTR optimization or considerations for real-world deployment scenarios.

Another promising direction is retrieval-augmented synthesis. In the text-to-image arena, Re-Imagen [5] retrieves relevant image–text pairs to guide synthesis, thereby improving the fidelity of generated images even for rare or complex entities. Similarly, Retrieval-Augmented Diffusion Models [2] propose a semi-parametric approach in which the diffusion model is conditioned on a set of nearest-neighbor images retrieved from an external database.

2.2 AIGC in E-commerce

Recent advances in AIGC have driven the development of innovative methods for e-commerce image synthesis, aiming to enhance business outcomes [11, 21, 38, 42, 44]. For instance, CG4CTR [44] is a diffusion-based creative generation pipeline that utilizes the diffusion model to alter image backgrounds, aiming to enhance CTR. This pipeline integrates a prompt model to generate personalized creatives and a reward model that evaluates multimodal features, thereby guiding the generation process. In addition, Fan et al. [11] developed an automated framework for generating product-image sequences in e-commerce, and Xue et al. [42] proposed a controllable, ID-preserved approach for accessory advertising image generation. Furthermore, the work by Ku et al. [21], specifically designed for furniture images, proposed a method called “copy-paste staging,” which involves retrieving staged backgrounds from similar products and integrating them into un-staged product images using a Generative Adversarial Network-based inpainting model. However, it lacks an integrated pipeline that includes a CTR prediction module, which limits its ability to assure business value. Additionally, this approach has not been deployed on a real-world business platform, therefore, its efficacy has not been validated on large-scale user data. A recent work by Chen et al. [6] leverages Multimodal Large Language Models (MLLM) fine-tuned through reinforcement learning with a CTR-driven reward model that captures user click preferences. While their approach achieves outstanding performance in various metrics, it incurs substantial computational costs, which may hinder rapid adaptation to evolving consumer preferences.

2.3 Click-Through Rate Prediction

Click-Through Rate (CTR) Prediction involves estimating the likelihood that users will click on a specific item, a task that is crucial for enhancing user experience and boosting revenue. Traditionally, prediction models [15, 34, 35, 46] have relied primarily on historical click data. Recent advancements have incorporated multimodal features such as visual and textual features to enhance prediction accuracy. For visual content, traditional methods relied on manually engineered visual features to estimate user engagement, and more recently, convolutional neural networks (CNN) have enabled automatic learning of hierarchical visual representations from raw images, capturing both low-level patterns and high-level semantic content for CTR predictions [4, 25, 43]. In parallel, large-scale pre-trained language models can generate dense textual embeddings that encapsulate semantic and contextual information from item titles or descriptions. By integrating these learned textual representations, modern CTR frameworks achieve substantially improved prediction accuracy [29, 40, 41]. Equally important is modeling the interactions among these modalities [14, 24, 37]. For example, AutoInt [37] integrates the attention mechanism within the CTR prediction framework to capture complex feature combinations, and MARN [24] uses adversarial training alongside attention to learn both modality-specific and modality-invariant representations. Moreover, HyperCTR [14] leverages multi-modal hypergraphs to capture complex user-item interactions across various modalities, effectively modeling both temporal and group-level user preferences for accurate CTR prediction.

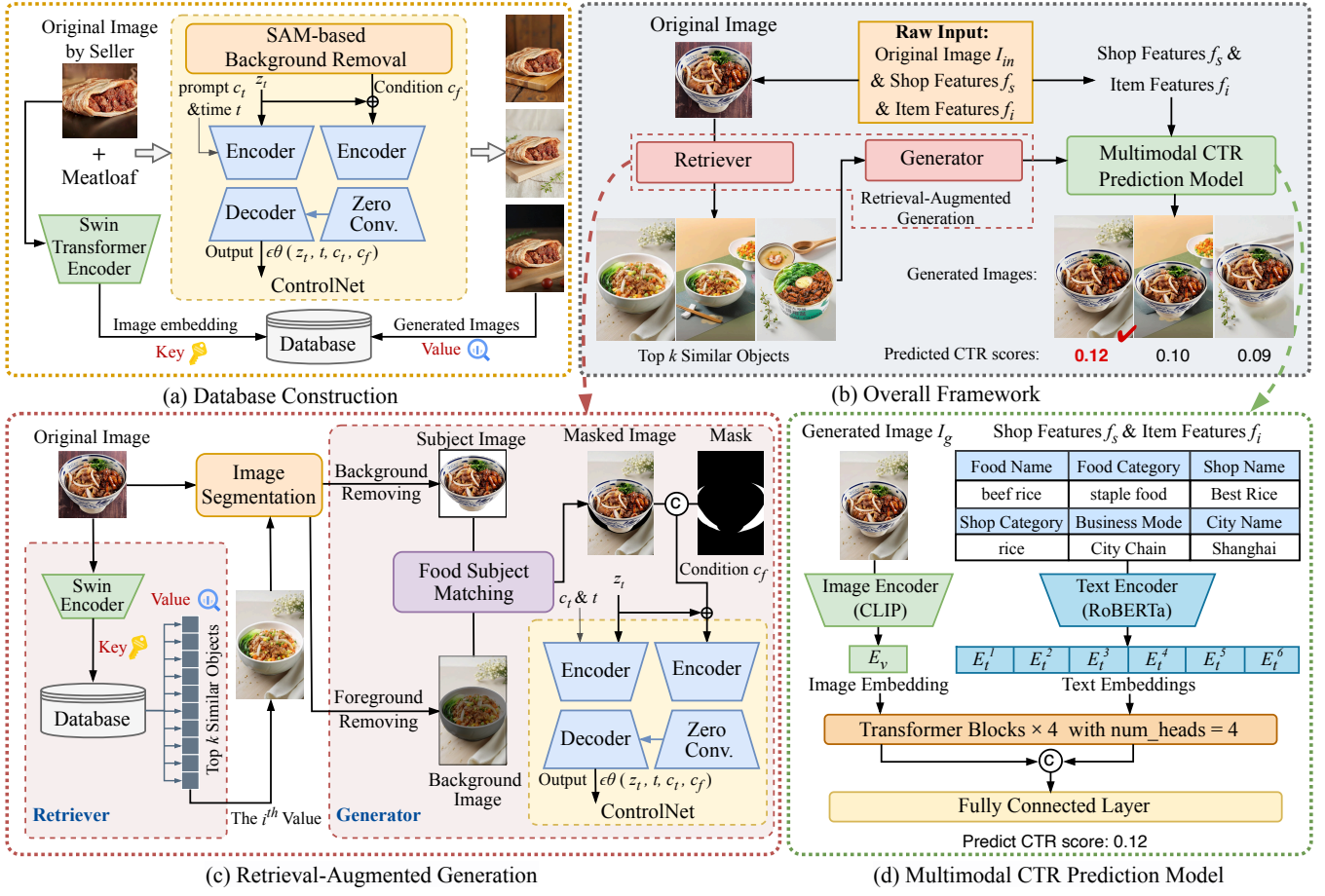


Figure 3: The architecture of our proposed FoRAGE pipeline. (a) is the process of database construction. (b) is the structure of the overall framework, which consists of (c), the retrieval-augmented generation module, and (d), the multimodal CTR prediction model. Notations: $\odot$ indicates concatenation, and $\oplus$ means add.

3 Methods

3.1 Overview

The architecture of our proposed **FoRAGE** pipeline is shown in Figure 3. Specifically, our method includes the following four main components, where Similar Image Retrieval and Hi-Fi Background Replacement serve as the retrieval step and generation step, respectively, forming a Retrieval-Augmented Generation process.

Database Construction. We first construct a retrieval database containing a large number of high-quality food images with visually enhanced backgrounds. For each food image, we isolate the food item using a SAM-based [20] background removal model. The resulting foreground, along with a corresponding text prompt, is then fed into a ControlNet architecture to generate 2-5 diverse backgrounds. Each food image with regenerated backgrounds is stored as a value in a database. The process is shown in Figure 3 (a).

Similar Image Retrieval. Given an input food image, we use an image encoder to extract its embedding and utilize an image retriever to search for the top- k background images in the retrieval

database that are similar to the subject of the input image. The process is shown in Figure 3 (c) Retriever.

Hi-Fi Background Replacement. We extract the food subject from the user-input image and extract the background from the retrieved image. Then the food subject and background are matched together. We use a pre-trained diffusion model to fill in the remaining vacancy and generate the final synthetic image. This process not only preserves the high quality of the background but also further enhances the naturalness and aesthetic appeal of the synthetic image through the generative capabilities of the diffusion model. The process is shown in Figure 3 (c) Generator.

Multimodal CTR Prediction. We train a multimodal CTR prediction model tasked with evaluating the k generated candidate images. This model comprehensively analyzes both visual and textual features to determine which image holds the greatest potential for attracting user clicks. By selecting the most appealing image, the model significantly improves the likelihood of achieving higher CTR, thereby optimizing the overall effectiveness of the image generation process for real-world deployment. The process is shown in Figure 3 (d).

3.2 Similar Image Retrieval

When merchants provide original images, we aim to retrieve similar high-quality images from our retrieval database. To construct the database, we employ a background generation method based on Stable Diffusion and ControlNet to create multiple distinct backgrounds for each subject. The text prompts are generated by an Alibaba internal Multimodal Large Language Model conditioned on the input image content. This approach enables us to generate various enhanced versions of each original image. These enhanced images not only retain the key information of the original subject but also possess more visually appealing backgrounds. Each enhanced image undergoes rigorous manual review before being stored in the database. This human-in-the-loop process is essential for ensuring high-quality and brand-safe outputs, which is critical in commercial applications. Finally, we store paired entries of original images and their approved enhancements in the retrieval database. The construction process of the retrieval database is shown in Figure 3 (a).

A pre-trained image encoder is used to encode the original image into an embedding stored in a vector database. We designate the embedding as the key and the higher-quality image as the value. When a new original merchant image is provided, we employ the same image encoder to maintain consistency in the representation space. We choose Swin transformer [27] pre-trained on the image classification task (ImageNet-1k) and fine-tuned on the Food101 dataset [3] as the image encoder due to its strong capability in capturing hierarchical representations of the image. Merchant-provided images typically highlight specific food items, with the main subject occupying a significant portion of the frame while the background remains relatively minor. Given this context, and considering that our image encoder is tailored for classification with a focus on the food subject, we encode the entire image in its original form without cropping or modification. This approach not only simplifies the processing workflow but also ensures the effectiveness and accuracy of the retrieval.

Following the acquisition of the embedding vector for the novel image, a similarity search is executed within the vector database. This search aims to identify the top- k most pertinent results in relation to the query image, predicated on the distance metric between the respective embedding vectors. Specifically, this model employs cosine similarity as the distance metric for the retrieval of analogous images.

3.3 Hi-Fi Background Replacement

When the retriever provides the top- k reference images, the generator replaces the background of the original image with the background of the reference image. This involves three steps: segmentation, positioning, and inpainting.

Segmentation. The segmentation process employs a deep learning based semantic segmentation model [18, 20, 22], which is pre-trained on a comprehensive dataset covering a broad range of image categories, endowing it with strong cross-category object recognition and segmentation capabilities. Specifically, when processing the retrieved reference images and the original merchant images, the model extracts the background devoid of the subject from the

reference images while segmenting the food item from the merchant images. This enables the precise separation of the target food from intricate backgrounds, providing an accurate foundation for subsequent image-processing tasks.

Positioning. Positioning the food subject onto the reference image can be achieved through two primary approaches: by feature-based matching using SIFT and FLANN, or via aligning the bounding boxes from the segmentation step.

The first method employs the Scale-Invariant Feature Transform (SIFT) [28] algorithm to detect and describe distinctive keypoints in both images. These descriptors are then matched using the Fast Library for Approximate Nearest Neighbors (FLANN) [30], which efficiently identifies corresponding features between the images. By computing a homography from these matches, we can accurately align and overlay the food subject onto the retrieved reference image.

Alternatively, the second approach utilizes the bounding boxes for the food subjects obtained from the previous segmentation step. By calculating the center coordinates of these bounding boxes, the food subject can be positioned at the corresponding location in the reference image. To ensure a natural fit, the food subject is also scaled to match the dimensions of the bounding box in the retrieved image.

Inpainting. During the inpainting stage, we employ a pre-trained diffusion model to fill in missing regions of the newly synthesized images. Based on the principles of diffusion probabilistic models, this process involves learning the complex distribution and semantic information of images by gradually adding noise to images and then reversing this process to denoise them. The diffusion model shows high sensitivity towards text prompts, especially when inpainting specific regions. However, it is often difficult to obtain accurate textual descriptions of the image's subject and background. Utilizing a multimodal model to extract image information may lead to incorrect prompts, potentially degrading the output quality. Therefore, the prompts used for inpainting are deliberately general—incorporating broad terms such as "neat," "tidy," or "food"—to reduce the risk of semantic misalignment.

To address potential residual noise caused by incomplete segmentation in background images, morphological dilation is applied to the mask of the removed subject before inpainting. This operation expands the mask boundaries, slightly enlarging the cutout area to ensure complete removal of the original subject and to provide a clean foundation for subsequent processing by the diffusion model.

Pre-trained models may arbitrarily expand salient objects during inpainting. To mitigate this, we employ a pre-trained diffusion model augmented with a ControlNet branch [10, 45], fine-tuned specifically for salient object outpainting tasks to avoid over-completion of objects. Moreover, although we dilate the mask to ensure complete removal of the original subject, we still aim to minimize the inpainting area as much as possible. Specifically, only the portions of the removed subject areas that are not covered by the newly positioned food subject are inpainted. By restricting such alterations, we preserve the integrity and authenticity of food images and mitigate potential compliance issues.

Notably, our approach significantly reduces the demand for the capabilities of the diffusion model. By minimizing the inpainting

Algorithm 1: Hi-Fi Background Replacement Algorithm

Input: Top- k reference images $R_1, R_2, \dots, R_k$, the image provided by the merchant O , the semantic segmentation model M_{seg} , the diffusion model M_{diff}

Result: Top- k high-quality images $I_1, I_2, \dots, I_k$

for $j = 1$ **to** k **do**

Segmentation:
 Through M_{seg} , obtain the food main bodies S_{R_j} and S_O , as well as the backgrounds B_{R_j} and B_O after deducting the main bodies from R_j and O respectively.

Positioning:
 SIFT & FLANN: Extract SIFT features from S_{R_j} and S_O , match them using FLANN, compute a transformation matrix, and warp S_O onto B_{R_j} , OR
 Bounding Box Matching: Use segmentation masks to obtain bounding boxes, then scale and translate S_O so its center aligns with the bounding-box center of S_{R_j} .

Inpainting:
 Dilate the mask of the deducted main body from B_{R_j} to obtain an expanded mask. Denote the corresponding region as I_{region} .
 Let c_t be the general text prompt (e.g., ‘neat’, ‘tidy’, ‘food’) and c_f is the masked image concatenated with the mask. Initialize z_T as a noisy version of I_{region} , where the noise is added according to the forward diffusion process $z_T = \sqrt{\bar{\alpha}_T} I_{region} + \sqrt{1 - \bar{\alpha}_T} \epsilon$, with $\bar{\alpha}_T = \prod_{s=1}^T \alpha_s$.
 Reverse Diffusion Loop:
for $t = T$ **down to** 1 **do**
 Predict the noise $\epsilon_\theta(z_t, t, c_t, c_f)$ using the Diffusion model M_{diff} Update
 $(z_{t-1} = \frac{1}{\sqrt{\alpha_t}}(z_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(z_t, t, c_t, c_f)))$
end
 Replace I_{region} with z_0 to obtain the high-quality image I_j ;
end

return $I_1, I_2, \dots, I_k$;

area, satisfactory results are achieved even under text prompts lacking specific information. And the diffusion model is not fine-tuned on food-related datasets, which further reduces the computational costs, making the method suitable for industrial deployment and easily adaptable to other domains.

3.4 Multimodal CTR Prediction

The CTR prediction model’s role is to predict the CTR of food images generated by the diffusion model from the same original image. Its structure is shown in Figure 3. The model’s inputs consist of six textual elements and one generated image. The textual inputs include the food name, food category, shop name, shop category, business mode, and city name. Each piece of text is integral to the CTR prediction model for distinct reasons. The food name enriches the visual data captured by the image encoder, offering an additional layer of context. The food category is essential as consumer

preferences often vary significantly across different categories. The shop name, displayed alongside the vertical image, contributes to CTR due to the influence of brand recognition. The shop category provides insight into the type of cuisine primarily offered, which can affect customer interest. The business mode, indicating whether establishments are chain stores or independent, for example, impacts consumer decisions. Finally, the city name is crucial, as it accounts for the diverse culinary tastes and preferences prevalent in different regions of China.

Text embeddings are generated from each of the textual features respectively using the pre-trained Chinese RoBERTa model [8]. We opted for Chinese RoBERTa with Whole Word Masking (WWM) due to its superior ability to handle Chinese textual data effectively. Building on the success of BERT [9], RoBERTa [26] enhances performance through optimized training strategies. WWM allows entire words to be masked during pre-training rather than individual characters. WWM is naturally aligned with Chinese Word Segmentation, which is crucial for maintaining semantic integrity, improving the model’s ability to grasp context and semantics.

Different from the Swin model in the retrieval phase, we opted for the pre-trained CLIP model [33] for extracting visual features in our CTR prediction model. While Swin is pre-trained primarily for classification tasks, excelling at identifying and categorizing objects within images, it may not effectively capture the subjective qualities related to visual appeal and user engagement. CLIP, on the other hand, is trained on a diverse range of internet images and corresponding text descriptions, enabling it to develop a richer understanding of aesthetics and contextual nuance [16]. This makes CLIP particularly well-suited for applications like ours, where the goal is to identify images that are visually appealing and more likely to attract user attention and clicks.

To seamlessly integrate textual and visual features within our model, we utilize multiple transformer encoder layers [39], which are pivotal in capturing complex self-attention representations. These layers allow the model to weigh the importance of different parts of the input sequences, thereby facilitating the learning of intricate relationships between textual and visual modalities. Once the self-attention features are obtained, they are concatenated into a single feature vector and fed into fully connected layers designed to interpret the aggregated features and predict the final CTR score.

$$E_v = CLIP(img), \quad E_t^i = RoBERTa(txt^i), \text{ for } i = 1, 2, \dots, 6 \quad (1)$$

$$\hat{E}_v, \hat{E}_t^1, \dots, \hat{E}_t^6 = transformers(E_v, E_t^1, \dots, E_t^6) \quad (2)$$

$$CTR_p = FC(Concat(\hat{E}_v, \hat{E}_t^1, \dots, \hat{E}_t^6)) \quad (3)$$

The real CTR value is computed using Equation (4). The loss function is formulated as the Mean Squared Error (MSE) between the real CTR (CTR_r) and the predicted CTR (CTR_p). The pre-trained RoBERTa and CLIP models’ parameters are frozen during training while allowing the transformer encoder layers to remain trainable.

$$CTR_r = \frac{\text{click}}{\text{exposure}} \quad (4)$$

4 Experiments

4.1 Configurations and Deployments

4.1.1 Implementation Details. To implement the retriever, we utilize ChromaDB version 0.6.2, an open-source multimodal vector database, with a Swin transformer (finetuned-food101) as the image encoder. Vector similarity is computed using cosine similarity. For the generator, we build upon a pre-trained model released by Yahoo (yahoo-inc/photo-background-generation), loading and running inference using Diffusers version 0.33.0.dev0. Both images and masks are padded and resized to 512x512 pixels, and the diffusion model performs 30 inference steps. Following Yahoo’s approach, the mask is also provided as control information to the model. The CTR prediction model utilizes the Chinese Roberta (wwm-ext) as the text encoder and CLIP (vit-base-patch32) as the image encoder. Multimodal feature fusion is achieved through a 4-layer transformer encoder architecture with 4 attention heads per layer.

Our experiments are conducted on hardware consisting of an NVIDIA H20 GPU and an Intel Xeon Platinum 8469C CPU. Under the specified experimental conditions, for each image, the retrieval stage requires approximately 0.3 seconds, the generation stage takes around 5 seconds, and the CTR prediction step incurs an additional 0.3 seconds. Such overall latency demonstrates that the proposed pipeline is sufficiently efficient for real-world deployment.

Our evaluation compares against a baseline method employing an Ele.me internal outpainting model for square-to-vertical image conversion. It is a ControlNet model fine-tuned on business-specific data, and currently represents the best-performing system deployed on the platform.

4.1.2 Deployment and Online Evaluation. The pipeline is deployed on the Ele.me platform with advertisements displayed in vertical poster formats across Ele.me APP, WeChat, and Alipay mini-programs. They target users on both the homepage and category pages. To ensure the randomness of our A/B test, each user ID is hashed and bucketed using a modulo operation. All experiment groups are allocated the same number of buckets and run for the same duration, ensuring fair and statistically sound comparisons. To evaluate the effectiveness of the pipeline, we utilized several key performance metrics: Click-Through Rate (CTR), Click-to-Conversion Rate (CTCVR), Cost, Revenue Per Mille (RPM), and Pay-Per-Click (PPC). Of these, CTR, which is defined in Equation (4), serves as the primary criterion, effectively gauging an advertisement’s ability to capture user attention. Here, Cost denotes the total expenditure incurred on the advertisement campaign by merchants. A higher cost typically translates to increased revenue for the delivery platform. The definitions of the other metrics are as follows:

$$\text{CTCVR} = \frac{\text{order}}{\text{exposure}} \quad (5)$$

$$\text{RPM} = \frac{\text{revenue}}{\text{exposure}} \times 1000 \quad (6)$$

$$\text{PPC} = \frac{\text{cost}}{\text{click}} \quad (7)$$

In our comparative analysis, we use images generated by the outpainting model as the baseline and calculate the percentage

improvement of the metrics to evaluate the efficacy using Equation (8), where M_o is our method’s measurement and M_b is the baseline measurement. All data presented in Table 1, Table 2, and Table 3 represent relative uplifts compared to the baseline.

$$\text{uplift} = \frac{M_o}{M_b} - 1 \quad (8)$$

4.2 Qualitative Analysis

Figure 4 illustrates a comparative analysis of images generated using our proposed pipeline against those produced via traditional outpainting techniques, with a marked improvement in CTR. This remarkable enhancement is primarily due to the introduction of aesthetically pleasing backgrounds that maintain consistency with the central food subjects. By swapping in more appropriate and visually appealing backgrounds, the generated images exhibit a considerable uplift in visual attractiveness. Note that the subject-matching algorithm employed in our pipeline ensures that the main subjects, or food items, are seamlessly integrated into images with optimized layouts. The generated images display subjects that are appropriately sized and positioned, significantly enhancing the overall composition and visual harmony of the images. Additionally, the retrieval process ensures that the new backgrounds not only complement but also stylistically align with the pre-existing content, thereby maintaining a cohesive and appealing visual narrative. Furthermore, our CTR prediction model plays an essential role in this process by effectively selecting images that exhibit the highest potential for visual appeal. As a result, the images generated through our pipeline consistently achieve higher CTR, demonstrating increased consumer engagement and interaction.

4.3 Quantitative Analysis

4.3.1 Online A/B Test. We conducted an online A/B test to compare their CTR to assess whether the images in our constructed retrieval database possess more visually appealing backgrounds compared to the baseline outpainting images. For each food item, an image with the same food subject is randomly selected from the retrieval database for the experiment. As detailed in Table 1, the overall CTR improvement is +3.10%, with a 90% confidence interval of (+1.70%, +4.49%). The online results indicate that images from the retrieval database consistently outperformed the baseline in terms of CTR, demonstrating a clear enhancement in background attractiveness. This ensures that the images generated from our pipeline possess more appealing backgrounds as well.

Table 1: Online A/B Test. All values in the table represent relative uplifts compared to the baseline internal outpainting model. ↑ indicates that higher is better.

Date	CTR ↑	CTCVR ↑	Cost ↑	RPM ↑
Day 1	+3.78%	+3.17%	+2.63%	+2.19%
Day 2	+1.66%	+5.42%	+0.36%	+0.30%
Day 3	+2.20%	-2.57%	+0.98%	+1.92%
Day 4	+4.74%	+3.07%	+3.53%	+4.92%
Overall	+3.10%	+2.08%	+1.87%	+2.32%

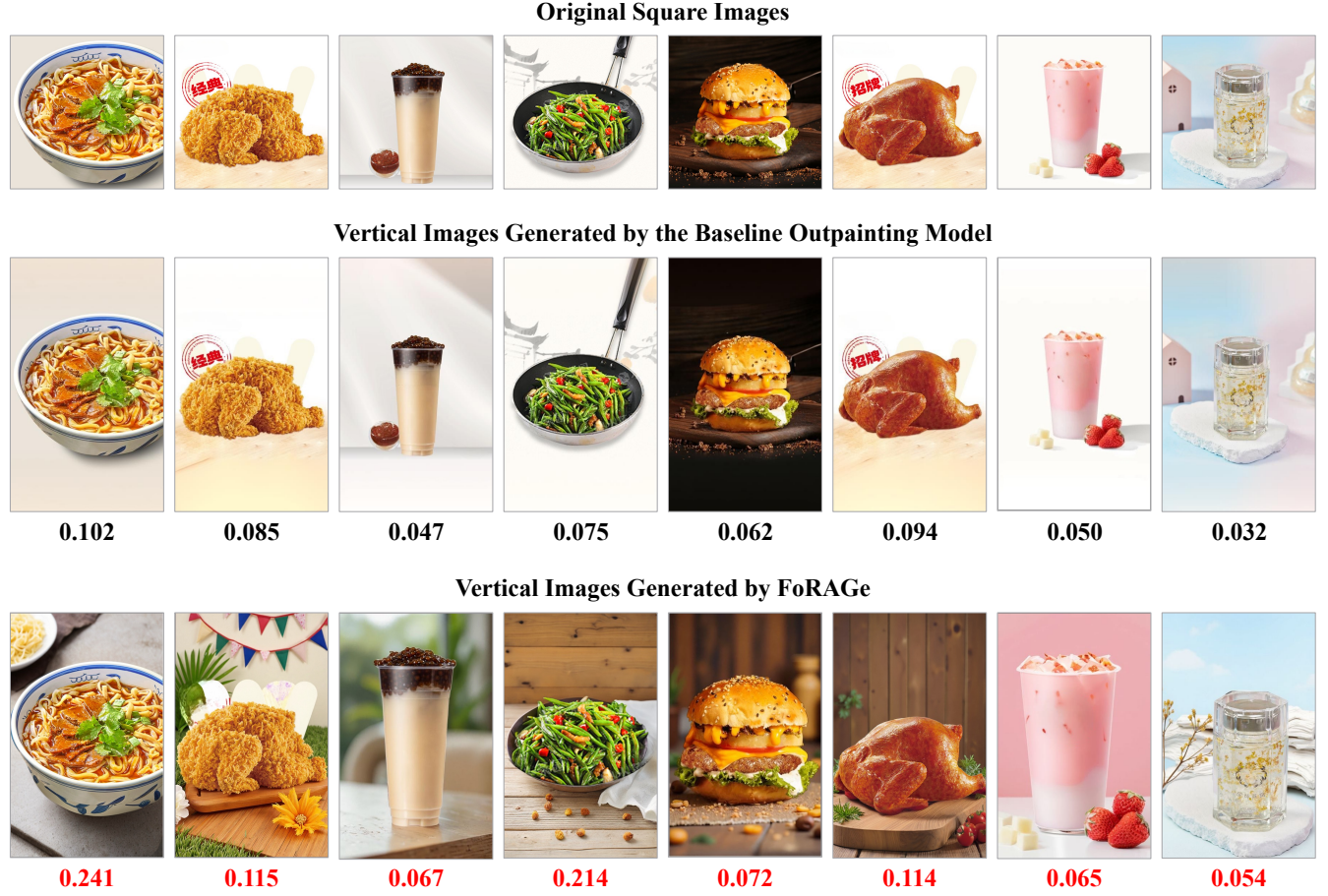


Figure 4: Examples showcasing original images, images with outpainting, and images generated using our pipeline. The CTR is displayed below each image, with higher values highlighted in red to indicate increased consumer engagement.

4.3.2 Ablation Study. Table 2 illustrates the results of the ablation study conducted on the impact of varying top- k values during the retrieval phase. A higher value of k indicates that more images are retrieved from the database, thus providing the CTR prediction model with a broader selection of generated images to evaluate. The data in Table 2 indicate that CTR uplift increases with larger values of k . This finding aligns with expectations, as a larger selection pool enables the CTR prediction model to more effectively identify and select the optimal image for display. Notably, with a top-10 selection, all food categories exhibit a substantial increase in CTR, particularly in Staple Food and Desserts & Beverages. This level of improvement is considered highly impactful in real-world applications, where even marginal gains can translate into significant business revenue.

Table 3 presents the results of the ablation study conducted on the CTR prediction model. In the experiments without the CTR prediction model, one of the k generated images was selected at random. In both top-5 and top-10 settings, the images selected by our CTR prediction model consistently outperform those chosen randomly. This underscores the critical role of the CTR prediction model in enhancing image selection quality.

5 Conclusion

In this paper, we introduce FoRAGE, a pipeline focusing on the generation of high-CTR food images for food delivery platforms. The pipeline encompasses three stages: retrieval, generation, and CTR prediction. Initially, a comprehensive retrieval database of high-quality background images is constructed. During the retrieval phase, top- k background exemplars featuring analogous food subjects are identified using embeddings extracted by a pre-trained image encoder. In the generation phase, these backgrounds are seamlessly integrated with the original food subjects using a Stable Diffusion-based inpainting model. Lastly, a multimodal CTR prediction model is utilized to select the optimal image from multiple candidates. Deployment on the Ele.me platform demonstrates significant improvements in key business metrics, including CTR, validating the effectiveness of the FoRAGE pipeline in enhancing consumer engagement. Our framework also enables practitioners in similar areas to quickly build a domain-specific retrieval database and a CTR prediction model using their business-specific data and plug it into our generation pipeline. This makes our solution broadly applicable to a wide range of domain-specific scenarios, beyond just food delivery.

Table 2: Ablation Study on Top- k of the Retrieval-Augmented Generation Module. All values in the table represent relative uplifts compared to the baseline internal outpainting model. The best results are in bold.

Top- k	Top-1				Top-3			
Food Categories	Staple Food	Snacks & Barbecue	Desserts & Beverages	Overall	Staple Food	Snacks & Barbecue	Desserts & Beverages	Overall
CTR $\uparrow$	+8.80%	+2.81%	+12.77%	+6.54%	+15.83%	-0.02%	+27.20%	+10.21%
RPM $\uparrow$	+9.72%	+38.81%	+49.36%	+23.83%	+14.02%	+9.31%	+52.88%	+15.48%
Cost $\uparrow$	+11.28%	+29.14%	+32.45%	+20.17%	+8.61%	-14.44%	+20.19%	+1.23%
PPC $\uparrow$	+0.85%	+35.01%	+32.45%	+16.23%	-1.57%	+9.33%	+20.19%	+4.78%

Top- k	Top-5				Top-10			
Food Categories	Staple Food	Snacks & Barbecue	Desserts & Beverages	Overall	Staple Food	Snacks & Barbecue	Desserts & Beverages	Overall
CTR $\uparrow$	+45.98%	-0.58%	+1.92%	+22.59%	+66.72%	+4.97%	+31.68%	+39.70%
RPM $\uparrow$	+16.01%	+14.45%	+12.12%	+14.22%	+67.12%	+61.23%	+83.48%	+66.24%
Cost $\uparrow$	+15.09%	+0.11%	+10.00%	+8.96%	+86.66%	+53.60%	+59.25%	+71.44%
PPC $\uparrow$	-20.53%	+15.12%	+10.00%	-6.83%	+0.24%	+53.60%	+39.34%	+19.00%

Table 3: Ablation Study on the Multimodal CTR Prediction Model (MCPM). All values in the table represent relative uplifts compared to the baseline internal outpainting model. The best results are in bold.

Top- k	Configuration	CTR $\uparrow$
Top-5	w/o MCPM	+10.25%
	w/ MCPM	+22.59%
Top-10	w/o MCPM	+4.87%
	w/ MCPM	+39.70%

6 Future Work

Moving forward, we plan to replace the current static prompts with a dynamic prompt generation module powered by a Multimodal Large Language Model (MLLM). This module will generate prompts conditioned on merchant-specific embeddings, user preference features, and the original product image. By incorporating individualized user profiles—such as browsing history, demographic data, and real-time context—the system can tailor prompts to each user’s unique interests and behavior patterns. The diffusion model will then use these adaptive, personalized prompts to generate more personalized and contextually relevant images. To fine-tune the diffusion model, we will leverage the CTR prediction model to differentiate between high-quality and low-quality image generations and apply reinforcement learning to optimize generation quality. Compared to traditional prompt models, MLLM-based approaches offer stronger priors, better generalization, higher diversity, and improved semantic alignment with user and product intent.

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